

Consider the example of heart attack predictions. If we observe an emergency room for several days, maybe, by coincidence, every admitted patient who is wearing a blue shirt is suffering from a heart attack and every admitted patient who is wearing a green shirt is healthy. A decision tree model that included shirt color in its prediction variables would pick up this pattern and use it as a branching variable because it has such high diagnostic accuracy in our observations. However, if we then use that decision tree to predict heart attacks in another hospital, or for some future day, we'll find that our predictions are often wrong, as many people in green shirts also suffer heart attacks and many people in blue shirts don't. The observations we used to build our decision tree are called *in-sample observations*, and the observations that we then test our model on, which are not part of our decision tree generation process, are called *out-of-sample observations*. Overfitting means that by zealously seeking low error rates in predictions of our in-sample observations, we have caused our decision tree model to have inordinately high error rates when predicting our out-of-sample observations.

Overfitting is a serious issue in all applications of machine learning, and it trips up even the best machine learning practitioners. To avoid it, we'll take an important step that will make our decision tree creation process better resemble the real-life prediction scenario.

Remember that real-life prediction is about the future, but when we build our decision tree we necessarily have data only from the past. We can't possibly get data from the future, so we'll split our dataset into two subsets: a *training set*, which we'll use only to build our decision tree, and a *test set*, which we'll use only to check the accuracy of our decision tree. Our test set is from the past, just like the rest of our data, but we treat it as if it's the future; we don't use it to create our decision tree (as if it hasn't happened yet), but we do use it—only after completely building the decision tree—to test the accuracy of our decision tree (as if we got it later in the future).

By doing this simple training/test split, we've made our decision tree generation process resemble the real-life problem of predicting the unknown future; the test set is like a simulated future. The error rate that we find on the test set gives us a reasonable expectation of the error rate we'll get from the actual future. We'll know that we're guilty of overfitting if the